

Reinforcement Learning for Music-Induced Emotion Recognition

Comparative Study: Q-Learning, SARSA, Double Q-Learning & Double SARSA

RL Project

Using the DEAP Dataset

April 16, 2026

Outline

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- 3 MDP Formulation
- 4 Reward Design
- 5 Algorithms
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Problem Statement

Objective

Use **Reinforcement Learning** to generate a personalized music playlist that transitions a listener's emotional state from any starting emotion (e.g. *anger*) to a target emotion (*happy*).

Key Challenges:

- Emotions are subjective and continuous
- Playlist must be short (6 clips)
- Must generalize across subjects
- Sparse reward signal

Approach:

- Model emotions on a 2D **Valence–Arousal** wheel
- Agent selects music clips as **actions**
- Minimize **angular error** to target
- Based on *Dutta et al. 2020*

Motivation & Background

- Music has a measurable effect on emotional states (EEG, self-reports)
- **Music therapy** and **affective computing** demand personalized playlists
- Prior work (Dutta et al. 2020) used Q-Learning achieving 19/32 convergence
- **Our contribution:**
 - Extend to **four** RL algorithms: Q-Learning, SARSA, Double Q-Learning, Double SARSA
 - Introduce **reward shaping** with stagnation penalty ($\lambda = -100$)
 - Add **terminal bonus** for near-target states
 - Achieve up to **32/32 convergence** (Double SARSA with shaping)

Dataset for Emotion Analysis using Physiological Signals (DEAP)

Koelstra et al., IEEE Trans. Affective Computing, 2012

Key Statistics:

- **32 participants**
- **40 music video clips** per participant
- Each clip rated on: **Valence** (1–9), **Arousal** (1–9), Dominance, Liking
- Self-reported emotional responses
- EEG and peripheral physiological signals recorded

Files Used:

- `participant_ratings.csv`
 - 1280 ratings (32×40)
 - Valence & Arousal columns
- `emotion_summary.csv`
 - 15 emotion labels
 - Mean Valence & Arousal per emotion

Emotion Grouping & Geneva Emotion Wheel

15 base emotions $\rightarrow$ 8 groups:

Group	Emotions
Happy	happy, joy
Pleasant	love, lovely, sentimental
Exciting	exciting, fun
Calm	mellow
Sad	sad, melancholy
Negative	depressing
Anger	hate, terrible
Fear	shock

Mapping to 2D Wheel:

- Valence & Arousal normalized to $[-1, 1]$
- Group center = mean of member emotions

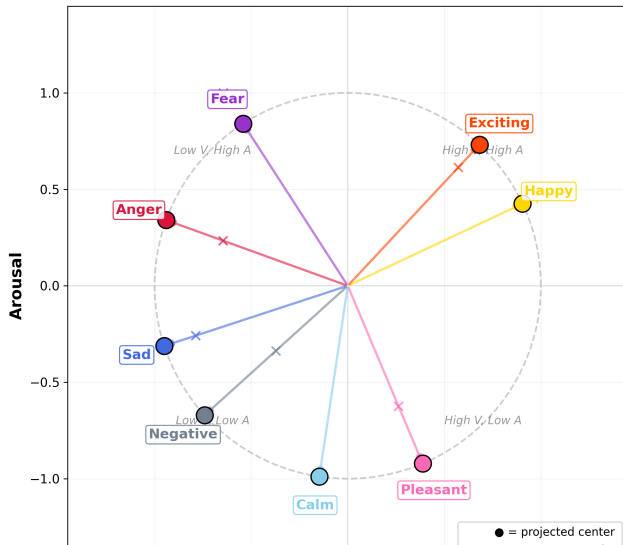
- Projected onto **unit circle**:

$$\vec{c}_g = \frac{\vec{v}_g}{\|\vec{v}_g\| + \epsilon}$$

- $\epsilon = 10^{-9}$ for numerical stability
- **Target**: “happy” emotion zone
- **Default start**: “anger” emotion zone

Grouped Geneva Emotion Wheel

Grouped Geneva Emotion Wheel (8 Clusters)



8 Emotion Zones on the Unit Circle:

Group	Angle
Happy	25.2°
Pleasant	-67.2°
Exciting	47.0°
Calm	-98.4°
Sad	-161.8°
Negative	-137.8°
Anger	160.1°
Fear	122.8°

The agent must navigate from **anger** (160°) to **happy** (25°) — spanning ~135° on the wheel.

MDP Formulation

State Space $\mathcal{S}$

- Continuous: 2D vector
 $\vec{s} = (v, a) \in [-1, 1]^2$
- Discretized: $20 \times 20 = \mathbf{400}$ grid cells
- Mapping: $s_{\text{idx}} = v_{\text{bin}} \times 20 + a_{\text{bin}}$

Transition Function

Additive transition with boundary projection:

$$\vec{s}' = \vec{s} + \vec{c}_a$$

$$\text{if } \|\vec{s}'\| > 1 : \quad \vec{s}' \leftarrow \frac{\vec{s}'}{\|\vec{s}'\|}$$

Action Space $\mathcal{A}$

- Each action = selecting a music clip
- **Global clip pool**: all training subjects' clips concatenated
- LOSO: $31 \times 40 = \mathbf{1240}$ clips per fold
- Clip represented as 2D vector (v_c, a_c)

Playlist Constraint

Agent selects exactly **6 clips** per episode (fixed trajectory length).

Reward Functions

1. Base Reward (Angular Criterion)

Let $\theta_1 = \angle(\vec{t}, \vec{c})$ and $\theta_2 = \angle(\vec{s}, \vec{c})$:

$$r_{\text{base}} = \begin{cases} 1.0 & \text{if } |\theta_1| + |\theta_2| \leq \pi \\ 0.0 & \text{otherwise} \end{cases}$$

where $\vec{t}$ = target, $\vec{c}$ = clip, $\vec{s}$ = state.

2. Terminal Bonus

$$r_{\text{terminal}} = \begin{cases} +10 & \text{if angular error} < 10 \\ 0 & \text{otherwise} \end{cases}$$

3. Reward Shaping (Stagnation Penalty)

$$\varphi(s, s') = \begin{cases} \lambda & \text{if } s_{\text{idx}} = s'_{\text{idx}} \\ 0 & \text{otherwise} \end{cases}$$

where $\lambda = -100$ (heavy penalty for no state change).

Total Reward

$$R = r_{\text{base}} + r_{\text{terminal}} + \varphi(s, s')$$

Reward shaping encourages the agent to **explore** and **avoid stagnation**.

Algorithm 1: Q-Learning (Off-Policy TD)

Update Rule

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \cdot \max_{a'} Q(s', a') + \varphi(s, s') - Q(s, a) \right]$$

Characteristics:

- **Off-policy:** uses greedy $\max_{a'}$ for value
- Single Q-table: $Q \in \mathbb{R}^{400 \times 1240}$
- Baseline from Dutta et al. 2020

Action Selection (ϵ -greedy):

$$a = \begin{cases} \text{random} & \text{with prob } \epsilon \\ \arg \max_a Q(s, a) & \text{otherwise} \end{cases}$$

Limitation: Prone to **maximization bias**.

Algorithm 2: SARSA (On-Policy TD)

Update Rule

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \cdot Q(s', a') + \varphi(s, s') - Q(s, a)]$$

where a' is the **actual next action** chosen by the ϵ -greedy policy.

Key Difference from Q-Learning:

- **On-policy:** evaluates the *same* policy it follows
- Uses $Q(s', a')$ instead of $\max_{a'} Q(s', a')$
- More conservative, accounts for exploration

Training Loop (SARSA-specific):

- 1 Initialize s , choose a (before loop)
- 2 Take action a , observe r, s'
- 3 Choose a' from s' (ϵ -greedy)
- 4 Update $Q(s, a)$ using $Q(s', a')$
- 5 $s \leftarrow s', a \leftarrow a'$

Algorithm 3: Double Q-Learning (van Hasselt 2010)

Update Rule (with probability 0.5 each)

Update Q_A : $Q_A(s, a) \leftarrow Q_A(s, a) + \alpha [r + \gamma \cdot Q_B(s', \arg \max_{a'} Q_A(s', a')) + \varphi - Q_A(s, a)]$

Update Q_B : (symmetric)

Key Idea:

- Two Q-tables: Q_A and Q_B
- **Decouples** action selection from evaluation
- Reduces **maximization bias**

Action Selection:

$$a = \arg \max_a [Q_A(s, a) + Q_B(s, a)]$$

Uses **combined** Q-tables for ϵ -greedy.

Algorithm 4: Double SARSA

Update Rule (with probability 0.5 each)

Update Q_A : $Q_A(s, a) \leftarrow Q_A(s, a) + \alpha [r + \gamma \cdot Q_B(s', a') + \varphi - Q_A(s, a)]$

Update Q_B : (symmetric, using Q_A for evaluation)

Combines benefits of:

- **Double learning:** two Q-tables reduce overestimation
- **On-policy (SARSA):** uses actual next action a'

Why it works best:

- No maximization bias
- No overoptimism (on-policy)
- Most stable convergence

Experimental Setup

LOSO Cross-Validation

- Train on 31 subjects, test on 1
- Repeated for all 32 subjects
- Global clip pool from training set

Hyperparameter Grid Search

10-fold CV, 500 episodes/fold:

- $\alpha \in \{0.05, 0.1, 0.2\}$
- $\gamma \in \{0.4, 0.6, 0.8\}$
- $\epsilon \in \{0.0, 0.05, 0.1, 0.2\}$

Training Configuration

Episodes: 1000 | **Playlist:** 6 clips | **Start:** “anger” (eval), all emotions (train) | **States:** $20 \times 20 = 400$ bins | **Actions:** $31 \times 40 = 1240$ clips/fold

Results: With Reward Shaping

Algorithm	Exact	< 30	< 60	Mean Error (Clip 6)
Q-Learning	29/32	25/32	30/32	21.6 ± 20.1
SARSA	22/32	22/32	28/32	30.2 ± 41.0
Double Q-Learning	30/32	27/32	31/32	17.1 ± 24.4
Double SARSA	32/32	29/32	32/32	13.4 ± 12.2
Paper (Q-Learn)	19/32	N/A	N/A	57.0 ± 2.8

Key Observations

- Double SARSA achieves **perfect 32/32** exact convergence
- All four algorithms significantly outperform the original paper (19/32)
- Mean error reduced from 57.0° (paper) to **13.4°** — a **76% reduction**

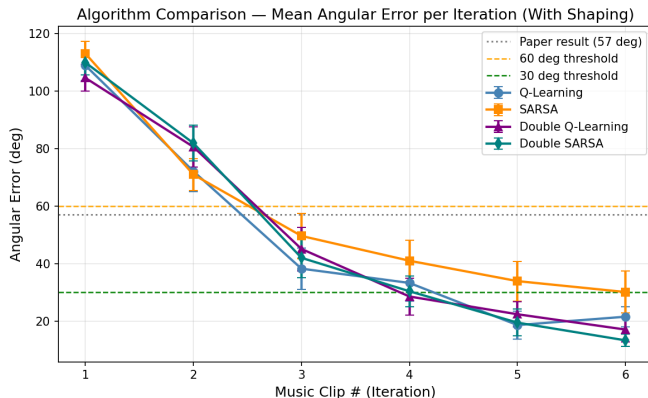
Results: Without Reward Shaping

Algorithm	Exact	< 30	< 60	Mean Error (Clip 6)
Q-Learning	24/32	25/32	27/32	31.1 ± 41.7
SARSA	20/32	20/32	28/32	32.3 ± 36.4
Double Q-Learning	20/32	21/32	23/32	45.4 ± 55.4
Double SARSA	27/32	23/32	27/32	31.6 ± 41.6
Paper (Q-Learn)	19/32	N/A	N/A	57.0 ± 2.8

Observations

- Without shaping, performance degrades significantly for all algorithms
- Double Q-Learning suffers the most: drops from 30/32 to 20/32, error jumps $17.1^\circ \rightarrow 45.4^\circ$
- Double SARSA remains the best even without shaping (27/32)
- High variance ($\pm 41\text{--}55^\circ$) indicates inconsistent convergence

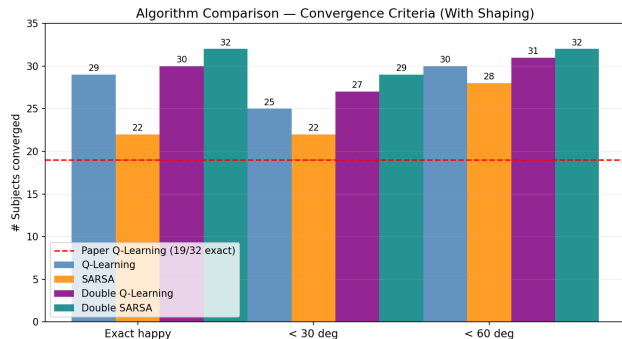
Angular Error Convergence — With Reward Shaping



Observations:

- All algorithms show monotonically decreasing error across the 6 clips
- **Double SARSA** (teal) converges fastest, reaching $\sim 13^\circ$ by clip 6
- **Double Q-Learning** (purple) closely follows at $\sim 17^\circ$
- **Q-Learning** (blue) settles at $\sim 22^\circ$
- **SARSA** (orange) is slowest at $\sim 30^\circ$

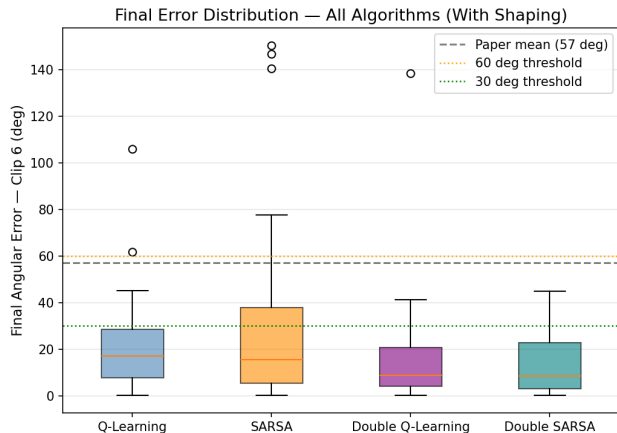
Convergence Criteria Comparison — With Reward Shaping



Observations:

- **Double SARSA** achieves the highest score in all three criteria
- Under the < 60 criterion, Double SARSA (32/32) and Double Q-Learning (31/32) near perfect
- All four algorithms surpass the paper baseline of 19/32 (red dashed line)
- SARSA lags behind in exact convergence (22/32) despite high < 60 count (28/32)

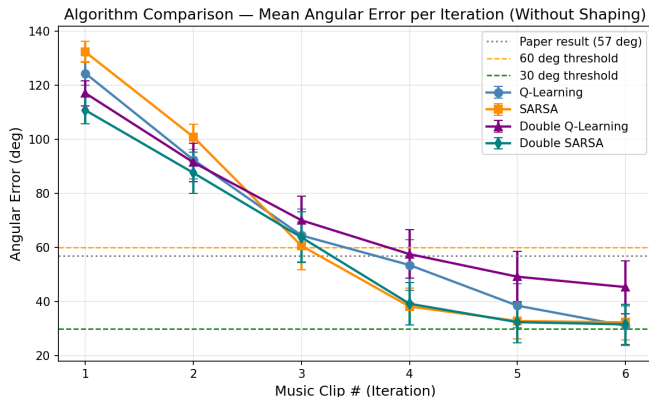
Final Error Distribution — With Reward Shaping



Observations:

- **Double SARSA** has the tightest box (lowest variance $\pm 12.2^\circ$) and lowest median
- Q-Learning and SARSA show high outliers (subjects stuck at $>100^\circ$)
- Double Q-Learning has a low median but some large outliers
- All medians are well below the paper mean of 57° (gray dashed)
- Double methods produce more **consistent** results across subjects

Angular Error Convergence — Without Reward Shaping

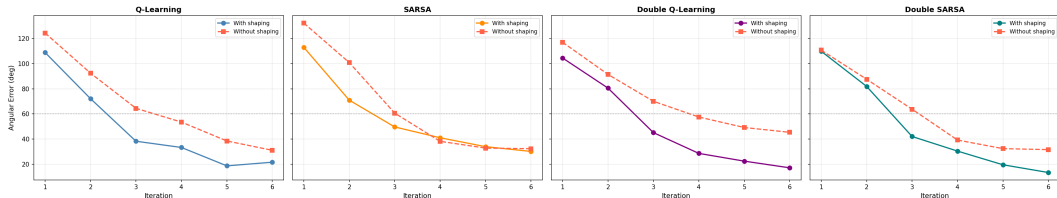


Observations:

- Without shaping, convergence is **slower and noisier** for all algorithms
- **Double Q-Learning** performs worst without shaping ($\sim 45^\circ$), showing it is most dependent on the stagnation penalty
- Q-Learning and Double SARSA perform similarly ($\sim 31^\circ$)
- Error curves are flatter — agents struggle to make progress in later clips
- SARSA is least affected by removing shaping

Effect of Reward Shaping — All Algorithms

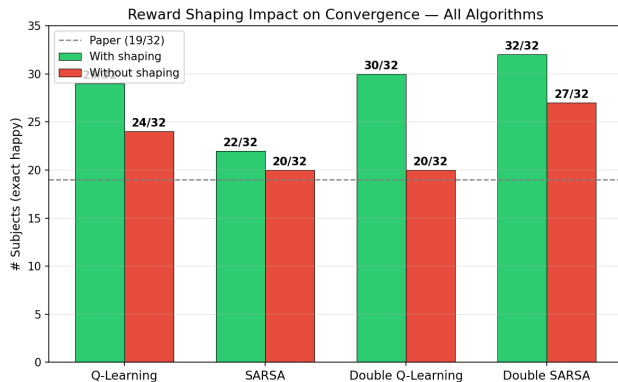
Reward Shaping Effect — All Algorithms



Key Finding

Reward shaping ($\varphi = -100$ on stagnation) consistently reduces angular error across all algorithms. **Double Q-Learning** shows the largest benefit ($45^\circ \rightarrow 17^\circ$), while **SARSA** is the most robust to its absence.

Reward Shaping Impact on Exact Convergence



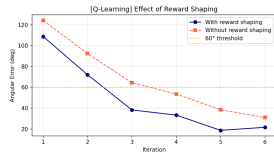
With vs Without Shaping:

Algorithm	With	W/o
Q-Learning	29	24
SARSA	22	20
Double QL	30	20
Double SARSA	32	27

Observations:

- Double Q-Learning has the **largest drop**: $30 \rightarrow 20$ (-33%)
- SARSA is least affected: $22 \rightarrow 20$ (-9%)
- Shaping is **critical** for Double methods

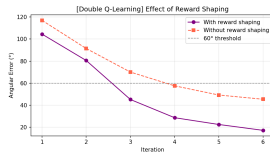
Per-Algorithm: Reward Shaping Comparison



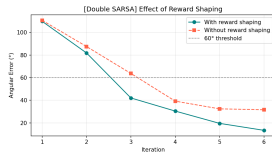
Q-Learning



SARSA



Double Q-Learning

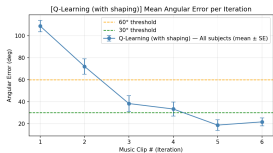


Double SARSA

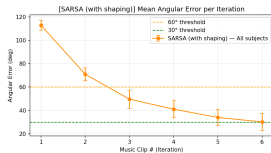
Observations

- In all plots, the solid line (with shaping) consistently lies **below** the dashed line (without)
- The gap is most dramatic for **Double Q-Learning** and **Double SARSA** from clip 3 onward
- SARSA shows the **smallest gap** — its on-policy nature provides inherent stability

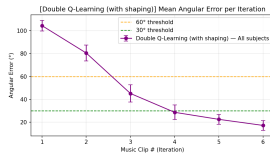
Angular Error Curves — Individual Algorithms (With Shaping)



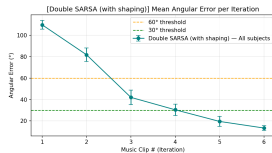
Q-Learning
 21.6 ± 20.1



SARSA
 30.2 ± 41.0



Double Q-Learn
 17.1 ± 24.4



Double SARSA
 13.4 ± 12.2

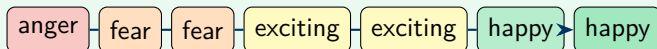
Observation

All algorithms show a clear downward trend with standard error bands. **Double SARSA** has the narrowest confidence band, confirming its **lowest variance** and most **reliable** performance.

Convergence Path Examples

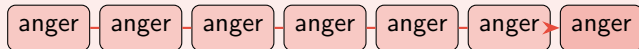
✓ Successful — Subject S16, Double SARSA w/ shaping (err = 21.1°)

err=21.1° | exact=v | <30°=v | <60°=v | start=anger



X Failed — Subject S07, Q-Learning w/ shaping (err = 133.5°)

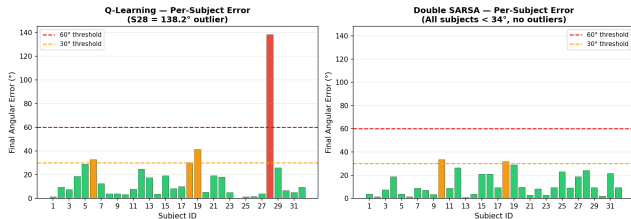
err=133.5° | exact=x | <30°=x | <60°=x | start=anger



Note: Double SARSA + shaping: **0 failures** (32/32). Failed paths appear only in single-table or no-shaping variants.

Why Does Convergence Fail?

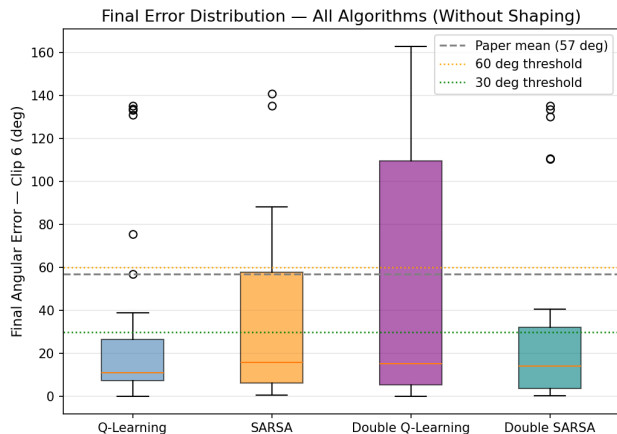
Subject Variability: Why Some Subjects Fail



Root Causes of Failure:

- 1 **Outlier subject ratings:** Subject S28 has atypical Valence-Arousal responses that deviate from the population mean, making the global clip pool ineffective.
- 2 **Q-table sparsity:** With 400 states $\times$ 1240 actions, most Q-table entries are never visited — the agent has no learned policy for rare states.
- 3 **Maximization bias (Q-Learning):** Overestimated Q-values cause the agent to repeatedly choose the same poor action, getting “stuck” in anger.

Final Error Distribution — Without Reward Shaping

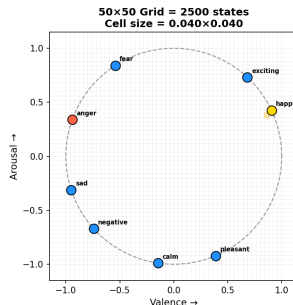
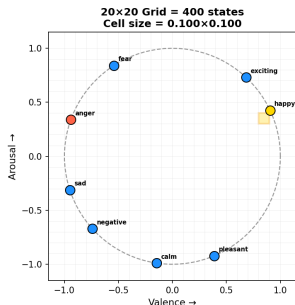


Observations:

- Without shaping, **all boxes expand** significantly
- **Double Q-Learning** has the widest spread ($\pm 55.4^\circ$) and highest median
- Multiple outliers near 130–140° indicate subjects completely “stuck”
- Even without shaping, Q-Learning and Double SARSA medians stay below 60°
- Variance is 2–3× higher than with shaping across the board

Limitation 1: Coarse State Discretization

State Space Discretization: Resolution Comparison



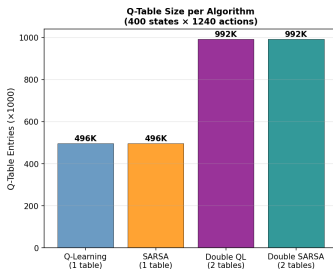
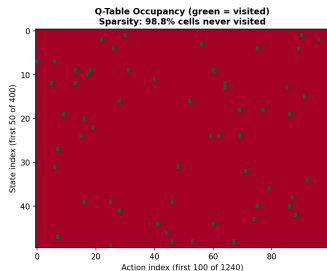
Problem:

- Continuous Valence-Arousal space discretized to $20 \times 20 = 400$ states
- Each cell spans 0.105×0.105 in normalized space
- Nearby but distinct emotions (e.g. *happy* vs. *exciting*) may map to the **same cell**
- Agent cannot distinguish fine-grained emotional differences

Impact:

A finer grid (e.g. $50 \times 50 = 2500$ states) improves resolution but exponentially

Limitation 2: Q-Table Sparsity & Scalability



Problem:

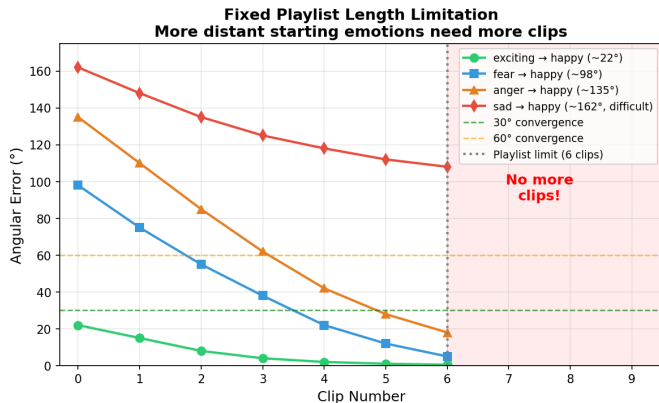
- Q-table: $400 \times 1240 = 496K$ entries
- Double methods:
 $2 \times 496K = 992K$ entries
- With $1000 \text{ episodes} \times 6 \text{ steps} = 6000$ updates, most cells **never visited**
- No generalization: similar states have independent Q-values

Consequence:

Tabular methods cannot scale to larger action spaces or continuous actions.

Deep RL (DQN) would enable function approximation and generalization

Limitation 3: Fixed Playlist Length



Problem:

- Playlist is fixed at **6 clips** regardless of starting emotion
- Nearby starts (exciting → happy, ~22°) converge in 2–3 clips — remaining clips are wasted
- Distant starts (anger → happy, ~135°) need all 6 clips and may still fall short
- Difficult subjects may need >6 clips to converge

Solution:

Adaptive playlist length with early stopping when angular error drops below threshold

Limitation 4: Additional Constraints

No Real-Time Feedback

- Uses **pre-collected** self-report ratings
- No EEG or physiological signal integration during playlist generation
- Cannot adapt to listener's *current* emotional state in real time
- Assumes static emotional response to each clip

Limited Dataset Scope

- Only **32 subjects** — small sample size
- Only **40 clips** per subject
- Western music stimuli only
- No cross-cultural validation
- Self-reported labels may not reflect true physiological emotion

No Personalization

All subjects share the **same global clip pool** and **same emotion wheel**. Individual differences in music perception are only captured through LOSO evaluation, not through personalized models.

Conclusion & Key Takeaways

Summary of Contributions

- 1 Extended Dutta et al. 2020 to **four RL algorithms**
- 2 Introduced **reward shaping** ($\lambda = -100$) and **terminal bonus** (+10)
- 3 Achieved **32/32 exact convergence** with Double SARSA (vs. 19/32 in paper)
- 4 Mean angular error: **13.4°** (vs. 57.0° in paper) — 76% reduction
- 5 Demonstrated **reward shaping is critical** for stable convergence

Best Algorithm

Double SARSA with reward shaping:

- 100% exact convergence
- Lowest mean error (13.4°)
- Lowest variance ($\pm 12.2^\circ$)

Future Work

- Deep RL (DQN, A3C)
- Real-time EEG feedback
- Larger datasets (DREAMER)
- User study for validation

Thank You!

Questions?

Code: `music_emotion_compare.py` | Dataset: DEAP

Reference: Dutta et al., “Music Emotion Recognition using EEG”, 2020